

2.5 | The Precise Definition of a Limit

Learning Objectives

- 2.5.1** Describe the epsilon-delta definition of a limit.
- 2.5.2** Apply the epsilon-delta definition to find the limit of a function.
- 2.5.3** Describe the epsilon-delta definitions of one-sided limits and infinite limits.
- 2.5.4** Use the epsilon-delta definition to prove the limit laws.

By now you have progressed from the very informal definition of a limit in the introduction of this chapter to the intuitive understanding of a limit. At this point, you should have a very strong intuitive sense of what the limit of a function means and how you can find it. In this section, we convert this intuitive idea of a limit into a formal definition using precise mathematical language. The formal definition of a limit is quite possibly one of the most challenging definitions you will encounter early in your study of calculus; however, it is well worth any effort you make to reconcile it with your intuitive notion of a limit. Understanding this definition is the key that opens the door to a better understanding of calculus.

Quantifying Closeness

Before stating the formal definition of a limit, we must introduce a few preliminary ideas. Recall that the distance between two points a and b on a number line is given by $|a - b|$.

- The statement $|f(x) - L| < \varepsilon$ may be interpreted as: *The distance between $f(x)$ and L is less than ε .*
- The statement $0 < |x - a| < \delta$ may be interpreted as: *$x \neq a$ and the distance between x and a is less than δ .*

It is also important to look at the following equivalences for absolute value:

- The statement $|f(x) - L| < \varepsilon$ is equivalent to the statement $L - \varepsilon < f(x) < L + \varepsilon$.
- The statement $0 < |x - a| < \delta$ is equivalent to the statement $a - \delta < x < a + \delta$ and $x \neq a$.

With these clarifications, we can state the formal **epsilon-delta definition of the limit**.

Definition

Let $f(x)$ be defined for all $x \neq a$ over an open interval containing a . Let L be a real number. Then

$$\lim_{x \rightarrow a} f(x) = L$$

if, for every $\varepsilon > 0$, there exists a $\delta > 0$, such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

This definition may seem rather complex from a mathematical point of view, but it becomes easier to understand if we break it down phrase by phrase. The statement itself involves something called a *universal quantifier* (for every $\varepsilon > 0$), an *existential quantifier* (there exists a $\delta > 0$), and, last, a *conditional statement* (if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$). Let's take a look at **Table 2.9**, which breaks down the definition and translates each part.

Definition	Translation
1. For every $\varepsilon > 0$,	1. For every positive distance ε from L ,
2. there exists a $\delta > 0$,	2. There is a positive distance δ from a ,
3. such that	3. such that
4. if $0 < x - a < \delta$, then $ f(x) - L < \varepsilon$.	4. if x is closer than δ to a and $x \neq a$, then $f(x)$ is closer than ε to L .

Table 2.9 Translation of the Epsilon-Delta Definition of the Limit

We can get a better handle on this definition by looking at the definition geometrically. **Figure 2.39** shows possible values of δ for various choices of $\varepsilon > 0$ for a given function $f(x)$, a number a , and a limit L at a . Notice that as we choose smaller values of ε (the distance between the function and the limit), we can always find a δ small enough so that if we have chosen an x value within δ of a , then the value of $f(x)$ is within ε of the limit L .

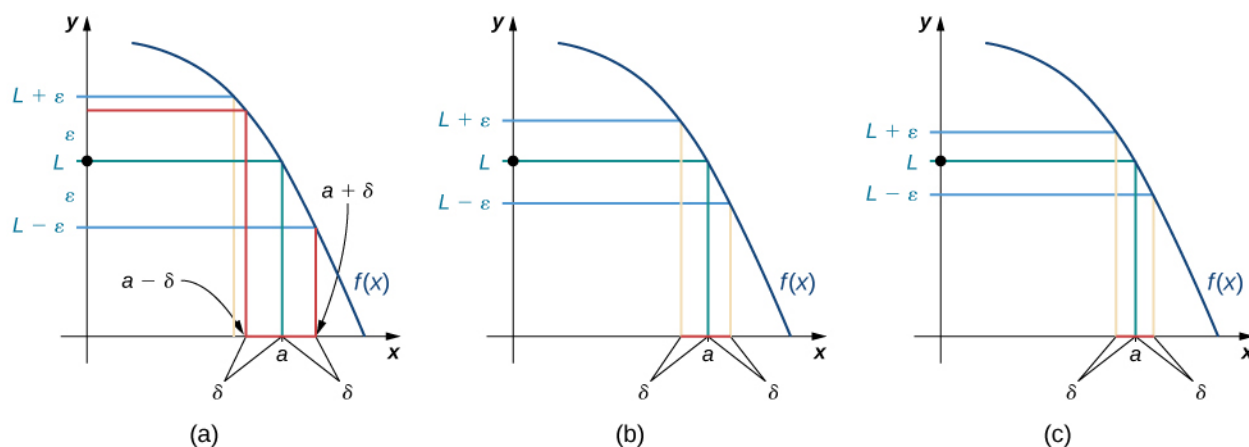


Figure 2.39 These graphs show possible values of δ , given successively smaller choices of ε .



Visit the following applet to experiment with finding values of δ for selected values of ε :

- **The epsilon-delta definition of limit** (http://www.openstax.org//20_epsilondelta)

Example 2.39 shows how you can use this definition to prove a statement about the limit of a specific function at a specified value.

Example 2.39

Proving a Statement about the Limit of a Specific Function

Prove that $\lim_{x \rightarrow 1} (2x + 1) = 3$.

Solution

Let $\varepsilon > 0$.

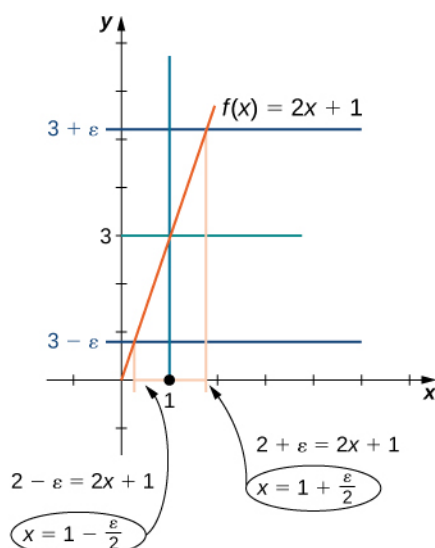
The first part of the definition begins “For every $\varepsilon > 0$.” This means we must prove that whatever follows is true no matter what positive value of ε is chosen. By stating “Let $\varepsilon > 0$,” we signal our intent to do so.

Choose $\delta = \frac{\varepsilon}{2}$.

The definition continues with “there exists a $\delta > 0$.” The phrase “there exists” in a mathematical statement is always a signal for a scavenger hunt. In other words, we must go and find δ . So, where exactly did $\delta = \varepsilon/2$ come from? There are two basic approaches to tracking down δ . One method is purely algebraic and the other is geometric.

We begin by tackling the problem from an algebraic point of view. Since ultimately we want $|(2x + 1) - 3| < \varepsilon$, we begin by manipulating this expression: $|(2x + 1) - 3| < \varepsilon$ is equivalent to $|2x - 2| < \varepsilon$, which in turn is equivalent to $|2||x - 1| < \varepsilon$. Last, this is equivalent to $|x - 1| < \varepsilon/2$. Thus, it would seem that $\delta = \varepsilon/2$ is appropriate.

We may also find δ through geometric methods. **Figure 2.40** demonstrates how this is done.



δ is the length of the smaller of the two distances marked in brown.

$$\begin{aligned}\delta &= \min \left\{ 1 + \frac{\varepsilon}{2} - 1, 1 - \left(1 - \frac{\varepsilon}{2} \right) \right\} \\ &= \min \left\{ \frac{\varepsilon}{2}, \frac{\varepsilon}{2} \right\} \\ &= \frac{\varepsilon}{2}\end{aligned}$$

Figure 2.40 This graph shows how we find δ geometrically.

Assume $0 < |x - 1| < \delta$. When δ has been chosen, our goal is to show that if $0 < |x - 1| < \delta$, then $|(2x + 1) - 3| < \varepsilon$. To prove any statement of the form “If this, then that,” we begin by assuming “this” and trying to get “that.”

Thus,

$$\begin{aligned}|(2x + 1) - 3| &= |2x - 2| && \text{property of absolute value} \\ &= |2(x - 1)| \\ &= |2||x - 1| && |2| = 2 \\ &= 2|x - 1| \\ &< 2 \cdot \delta && \text{here's where we use the assumption that } 0 < |x - 1| < \delta \\ &= 2 \cdot \frac{\varepsilon}{2} = \varepsilon && \text{here's where we use our choice of } \delta = \varepsilon/2\end{aligned}$$

Analysis

In this part of the proof, we started with $|(2x + 1) - 3|$ and used our assumption $0 < |x - 1| < \delta$ in a key part of the chain of inequalities to get $|(2x + 1) - 3|$ to be less than ε . We could just as easily have manipulated the assumed inequality $0 < |x - 1| < \delta$ to arrive at $|(2x + 1) - 3| < \varepsilon$ as follows:

$$\begin{aligned}
 0 < |x - 1| < \delta &\Rightarrow |x - 1| < \delta \\
 &\Rightarrow -\delta < x - 1 < \delta \\
 &\Rightarrow -\frac{\varepsilon}{2} < x - 1 < \frac{\varepsilon}{2} \\
 &\Rightarrow -\varepsilon < 2x - 2 < \varepsilon \\
 &\Rightarrow -\varepsilon < 2x - 2 < \varepsilon \\
 &\Rightarrow |2x - 2| < \varepsilon \\
 &\Rightarrow |(2x + 1) - 3| < \varepsilon.
 \end{aligned}$$

Therefore, $\lim_{x \rightarrow 1} (2x + 1) = 3$. (Having completed the proof, we state what we have accomplished.)

After removing all the remarks, here is a final version of the proof:

Let $\varepsilon > 0$.

Choose $\delta = \varepsilon/2$.

Assume $0 < |x - 1| < \delta$.

Thus,

$$\begin{aligned}
 |(2x + 1) - 3| &= |2x - 2| \\
 &= |2(x - 1)| \\
 &= |2||x - 1| \\
 &= 2|x - 1| \\
 &< 2 \cdot \delta \\
 &= 2 \cdot \frac{\varepsilon}{2} \\
 &= \varepsilon.
 \end{aligned}$$

Therefore, $\lim_{x \rightarrow 1} (2x + 1) = 3$.

The following Problem-Solving Strategy summarizes the type of proof we worked out in **Example 2.39**.

Problem-Solving Strategy: Proving That $\lim_{x \rightarrow a} f(x) = L$ for a Specific Function $f(x)$

1. Let's begin the proof with the following statement: Let $\varepsilon > 0$.
2. Next, we need to obtain a value for δ . After we have obtained this value, we make the following statement, filling in the blank with our choice of δ : Choose $\delta = \underline{\hspace{2cm}}$.
3. The next statement in the proof should be (at this point, we fill in our given value for a): Assume $0 < |x - a| < \delta$.
4. Next, based on this assumption, we need to show that $|f(x) - L| < \varepsilon$, where $f(x)$ and L are our function $f(x)$ and our limit L . At some point, we need to use $0 < |x - a| < \delta$.
5. We conclude our proof with the statement: Therefore, $\lim_{x \rightarrow a} f(x) = L$.

Example 2.40

Proving a Statement about a Limit

Complete the proof that $\lim_{x \rightarrow -1} (4x + 1) = -3$ by filling in the blanks.

Let _____.

Choose $\delta =$ _____.

Assume $0 < |x - \text{_____}| < \delta$.

Thus, $|\text{_____} - \text{_____}| = \text{_____} \varepsilon$.

Solution

We begin by filling in the blanks where the choices are specified by the definition. Thus, we have

Let $\varepsilon > 0$.

Choose $\delta =$ _____.

Assume $0 < |x - (-1)| < \delta$. (or equivalently, $0 < |x + 1| < \delta$.)

Thus, $|(4x + 1) - (-3)| = |4x + 4| = |4||x + 1| < 4\delta \text{_____} \varepsilon$.

Focusing on the final line of the proof, we see that we should choose $\delta = \frac{\varepsilon}{4}$.

We now complete the final write-up of the proof:

Let $\varepsilon > 0$.

Choose $\delta = \frac{\varepsilon}{4}$.

Assume $0 < |x - (-1)| < \delta$ (or equivalently, $0 < |x + 1| < \delta$.)

Thus, $|(4x + 1) - (-3)| = |4x + 4| = |4||x + 1| < 4\delta = 4(\varepsilon/4) = \varepsilon$.



2.27 Complete the proof that $\lim_{x \rightarrow 2} (3x - 2) = 4$ by filling in the blanks.

Let _____.

Choose $\delta =$ _____.

Assume $0 < |x - \text{_____}| < \text{_____}$.

Thus,

$|\text{_____} - \text{_____}| = \text{_____} \varepsilon$.

Therefore, $\lim_{x \rightarrow 2} (3x - 2) = 4$.

In **Example 2.39** and **Example 2.40**, the proofs were fairly straightforward, since the functions with which we were working were linear. In **Example 2.41**, we see how to modify the proof to accommodate a nonlinear function.

Example 2.41

Proving a Statement about the Limit of a Specific Function (Geometric Approach)

Prove that $\lim_{x \rightarrow 2} x^2 = 4$.

Solution

1. Let $\varepsilon > 0$. The first part of the definition begins “For every $\varepsilon > 0$,” so we must prove that whatever follows is true no matter what positive value of ε is chosen. By stating “Let $\varepsilon > 0$,” we signal our intent to do so.
2. Without loss of generality, assume $\varepsilon \leq 4$. Two questions present themselves: Why do we want $\varepsilon \leq 4$ and why is it okay to make this assumption? In answer to the first question: Later on, in the process of solving for δ , we will discover that δ involves the quantity $\sqrt{4 - \varepsilon}$. Consequently, we need $\varepsilon \leq 4$. In answer to the second question: If we can find $\delta > 0$ that “works” for $\varepsilon \leq 4$, then it will “work” for any $\varepsilon > 4$ as well. Keep in mind that, although it is always okay to put an upper bound on ε , it is never okay to put a lower bound (other than zero) on ε .
3. Choose $\delta = \min\{2 - \sqrt{4 - \varepsilon}, \sqrt{4 + \varepsilon} - 2\}$. **Figure 2.41** shows how we made this choice of δ .

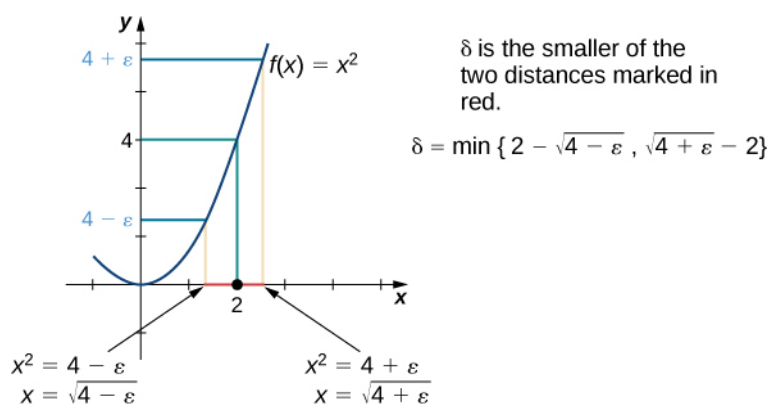


Figure 2.41 This graph shows how we find δ geometrically for a given ε for the proof in **Example 2.41**.

4. We must show: If $0 < |x - 2| < \delta$, then $|x^2 - 4| < \varepsilon$, so we must begin by assuming

$$0 < |x - 2| < \delta.$$

We don't really need $0 < |x - 2|$ (in other words, $x \neq 2$) for this proof. Since $0 < |x - 2| < \delta \Rightarrow |x - 2| < \delta$, it is okay to drop $0 < |x - 2|$.

$$|x - 2| < \delta.$$

Hence,

$$-\delta < x - 2 < \delta.$$

Recall that $\delta = \min\{2 - \sqrt{4 - \varepsilon}, \sqrt{4 + \varepsilon} - 2\}$. Thus, $\delta \leq 2 - \sqrt{4 - \varepsilon}$ and consequently $-(2 - \sqrt{4 - \varepsilon}) \leq -\delta$. We also use $\delta \leq \sqrt{4 + \varepsilon} - 2$ here. We might ask at this point: Why did we substitute $2 - \sqrt{4 - \varepsilon}$ for δ on the left-hand side of the inequality and $\sqrt{4 + \varepsilon} - 2$ on the right-hand side of the inequality? If we look at **Figure 2.41**, we see that $2 - \sqrt{4 - \varepsilon}$ corresponds to the distance on

the left of 2 on the x -axis and $\sqrt{4 + \varepsilon} - 2$ corresponds to the distance on the right. Thus,

$$-(2 - \sqrt{4 - \varepsilon}) \leq -\delta < x - 2 < \delta \leq \sqrt{4 + \varepsilon} - 2.$$

We simplify the expression on the left:

$$-2 + \sqrt{4 - \varepsilon} < x - 2 < \sqrt{4 + \varepsilon} - 2.$$

Then, we add 2 to all parts of the inequality:

$$\sqrt{4 - \varepsilon} < x < \sqrt{4 + \varepsilon}.$$

We square all parts of the inequality. It is okay to do so, since all parts of the inequality are positive:

$$4 - \varepsilon < x^2 < 4 + \varepsilon.$$

We subtract 4 from all parts of the inequality:

$$-\varepsilon < x^2 - 4 < \varepsilon.$$

Last,

$$|x^2 - 4| < \varepsilon.$$

5. Therefore,

$$\lim_{x \rightarrow 2} x^2 = 4.$$



2.28 Find δ corresponding to $\varepsilon > 0$ for a proof that $\lim_{x \rightarrow 9} \sqrt{x} = 3$.

The geometric approach to proving that the limit of a function takes on a specific value works quite well for some functions. Also, the insight into the formal definition of the limit that this method provides is invaluable. However, we may also approach limit proofs from a purely algebraic point of view. In many cases, an algebraic approach may not only provide us with additional insight into the definition, it may prove to be simpler as well. Furthermore, an algebraic approach is the primary tool used in proofs of statements about limits. For **Example 2.42**, we take on a purely algebraic approach.

Example 2.42

Proving a Statement about the Limit of a Specific Function (Algebraic Approach)

Prove that $\lim_{x \rightarrow -1} (x^2 - 2x + 3) = 6$.

Solution

Let's use our outline from the Problem-Solving Strategy:

1. Let $\varepsilon > 0$.
2. Choose $\delta = \min\{1, \varepsilon/5\}$. This choice of δ may appear odd at first glance, but it was obtained by

taking a look at our ultimate desired inequality: $|(x^2 - 2x + 3) - 6| < \varepsilon$. This inequality is equivalent to $|x + 1| \cdot |x - 3| < \varepsilon$. At this point, the temptation simply to choose $\delta = \frac{\varepsilon}{x - 3}$ is very strong.

Unfortunately, our choice of δ must depend on ε only and no other variable. If we can replace $|x - 3|$ by a numerical value, our problem can be resolved. This is the place where assuming $\delta \leq 1$ comes into play. The choice of $\delta \leq 1$ here is arbitrary. We could have just as easily used any other positive number. In some proofs, greater care in this choice may be necessary. Now, since $\delta \leq 1$ and $|x + 1| < \delta \leq 1$, we are able to show that $|x - 3| < 5$. Consequently, $|x + 1| \cdot |x - 3| < |x + 1| \cdot 5$. At this point we realize that we also need $\delta \leq \varepsilon/5$. Thus, we choose $\delta = \min\{1, \varepsilon/5\}$.

3. Assume $0 < |x + 1| < \delta$. Thus,

$$|x + 1| < 1 \text{ and } |x + 1| < \frac{\varepsilon}{5}.$$

Since $|x + 1| < 1$, we may conclude that $-1 < x + 1 < 1$. Thus, by subtracting 4 from all parts of the inequality, we obtain $-5 < x - 3 < -1$. Consequently, $|x - 3| < 5$. This gives us

$$|(x^2 - 2x + 3) - 6| = |x + 1| \cdot |x - 3| < \frac{\varepsilon}{5} \cdot 5 = \varepsilon.$$

Therefore,

$$\lim_{x \rightarrow -1} (x^2 - 2x + 3) = 6.$$



2.29 Complete the proof that $\lim_{x \rightarrow 1} x^2 = 1$.

Let $\varepsilon > 0$; choose $\delta = \min\{1, \varepsilon/3\}$; assume $0 < |x - 1| < \delta$.

Since $|x - 1| < 1$, we may conclude that $-1 < x - 1 < 1$. Thus, $1 < x + 1 < 3$. Hence, $|x + 1| < 3$.

You will find that, in general, the more complex a function, the more likely it is that the algebraic approach is the easiest to apply. The algebraic approach is also more useful in proving statements about limits.

Proving Limit Laws

We now demonstrate how to use the epsilon-delta definition of a limit to construct a rigorous proof of one of the limit laws. The **triangle inequality** is used at a key point of the proof, so we first review this key property of absolute value.

Definition

The **triangle inequality** states that if a and b are any real numbers, then $|a + b| \leq |a| + |b|$.

Proof

We prove the following limit law: If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$.

Let $\varepsilon > 0$.

Choose $\delta_1 > 0$ so that if $0 < |x - a| < \delta_1$, then $|f(x) - L| < \varepsilon/2$.

Choose $\delta_2 > 0$ so that if $0 < |x - a| < \delta_2$, then $|g(x) - M| < \varepsilon/2$.

Choose $\delta = \min\{\delta_1, \delta_2\}$.

Assume $0 < |x - a| < \delta$.

Thus,

$$0 < |x - a| < \delta_1 \text{ and } 0 < |x - a| < \delta_2.$$

Hence,

$$\begin{aligned} |(f(x) + g(x)) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

□

We now explore what it means for a limit not to exist. The limit $\lim_{x \rightarrow a} f(x)$ does not exist if there is no real number L for which $\lim_{x \rightarrow a} f(x) = L$. Thus, for all real numbers L , $\lim_{x \rightarrow a} f(x) \neq L$. To understand what this means, we look at each part of the definition of $\lim_{x \rightarrow a} f(x) = L$ together with its opposite. A translation of the definition is given in **Table 2.10**.

Definition	Opposite
1. For every $\varepsilon > 0$,	1. There exists $\varepsilon > 0$ so that
2. there exists a $\delta > 0$, so that	2. for every $\delta > 0$,
3. if $0 < x - a < \delta$, then $ f(x) - L < \varepsilon$.	3. There is an x satisfying $0 < x - a < \delta$ so that $ f(x) - L \geq \varepsilon$.

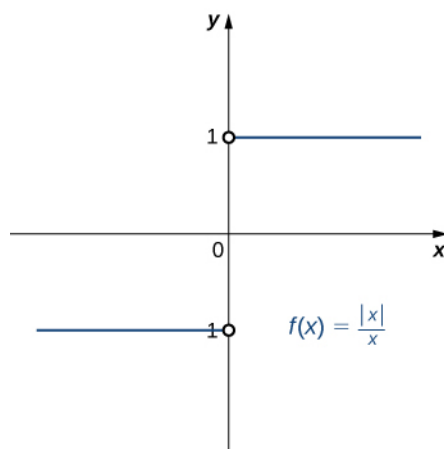
Table 2.10 Translation of the Definition of $\lim_{x \rightarrow a} f(x) = L$ and its Opposite

Finally, we may state what it means for a limit not to exist. The limit $\lim_{x \rightarrow a} f(x)$ does not exist if for every real number L , there exists a real number $\varepsilon > 0$ so that for all $\delta > 0$, there is an x satisfying $0 < |x - a| < \delta$, so that $|f(x) - L| \geq \varepsilon$. Let's apply this in **Example 2.43** to show that a limit does not exist.

Example 2.43

Showing That a Limit Does Not Exist

Show that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist. The graph of $f(x) = |x|/x$ is shown here:



Solution

Suppose that L is a candidate for a limit. Choose $\varepsilon = 1/2$.

Let $\delta > 0$. Either $L \geq 0$ or $L < 0$. If $L \geq 0$, then let $x = -\delta/2$. Thus,

$$|x - 0| = \left| -\frac{\delta}{2} - 0 \right| = \frac{\delta}{2} < \delta$$

and

$$\left| \frac{-\delta/2}{-\delta/2} - L \right| = |-1 - L| = L + 1 \geq 1 > \frac{1}{2} = \varepsilon.$$

On the other hand, if $L < 0$, then let $x = \delta/2$. Thus,

$$|x - 0| = \left| \frac{\delta}{2} - 0 \right| = \frac{\delta}{2} < \delta$$

and

$$\left| \frac{\delta/2}{\delta/2} - L \right| = |1 - L| = |L| + 1 \geq 1 > \frac{1}{2} = \varepsilon.$$

Thus, for any value of L , $\lim_{x \rightarrow 0} \frac{|x|}{x} \neq L$.

One-Sided and Infinite Limits

Just as we first gained an intuitive understanding of limits and then moved on to a more rigorous definition of a limit, we now revisit one-sided limits. To do this, we modify the epsilon-delta definition of a limit to give formal epsilon-delta definitions for limits from the right and left at a point. These definitions only require slight modifications from the definition of the limit. In the definition of the limit from the right, the inequality $0 < x - a < \delta$ replaces $0 < |x - a| < \delta$, which ensures that we only consider values of x that are greater than (to the right of) a . Similarly, in the definition of the limit from the left, the inequality $-\delta < x - a < 0$ replaces $0 < |x - a| < \delta$, which ensures that we only consider values of x that are less than (to the left of) a .

Definition

Limit from the Right: Let $f(x)$ be defined over an open interval of the form (a, b) where $a < b$. Then,

$$\lim_{x \rightarrow a^+} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < x - a < \delta$, then $|f(x) - L| < \varepsilon$.

Limit from the Left: Let $f(x)$ be defined over an open interval of the form (b, c) where $b < c$. Then,

$$\lim_{x \rightarrow a^-} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $-\delta < x - a < 0$, then $|f(x) - L| < \varepsilon$.

Example 2.44

Proving a Statement about a Limit From the Right

Prove that $\lim_{x \rightarrow 4^+} \sqrt{x-4} = 0$.

Solution

Let $\varepsilon > 0$.

Choose $\delta = \varepsilon^2$. Since we ultimately want $|\sqrt{x-4} - 0| < \varepsilon$, we manipulate this inequality to get $\sqrt{x-4} < \varepsilon$ or, equivalently, $0 < x - 4 < \varepsilon^2$, making $\delta = \varepsilon^2$ a clear choice. We may also determine δ geometrically, as shown in **Figure 2.42**.

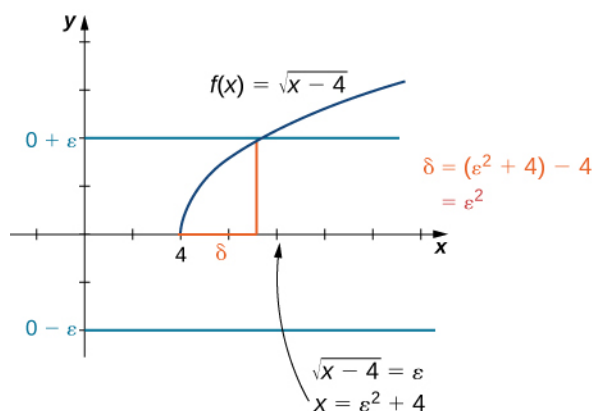


Figure 2.42 This graph shows how we find δ for the proof in **Example 2.44**.

Assume $0 < x - 4 < \delta$. Thus, $0 < x - 4 < \varepsilon^2$. Hence, $0 < \sqrt{x-4} < \varepsilon$. Finally, $|\sqrt{x-4} - 0| < \varepsilon$.

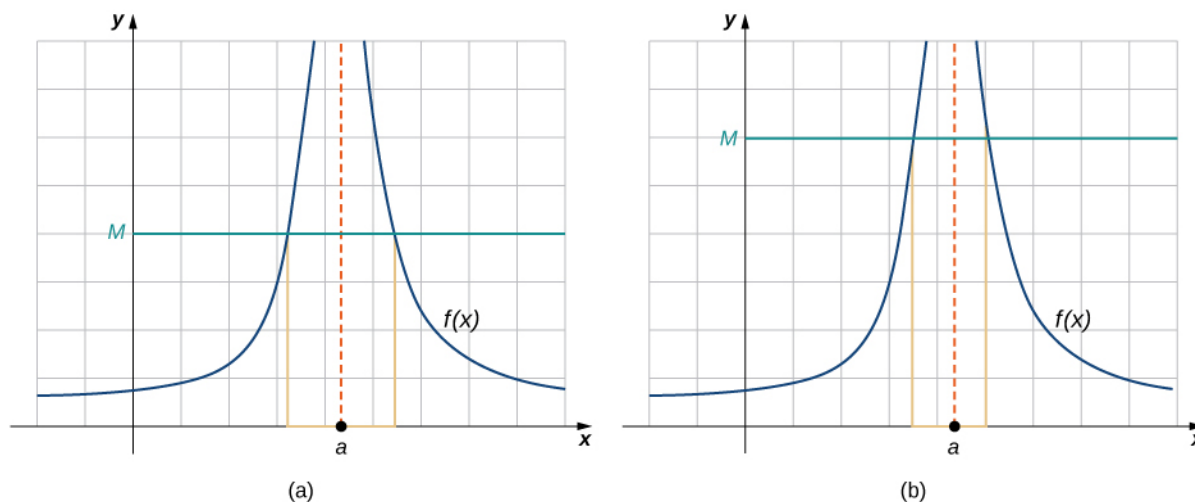
Therefore, $\lim_{x \rightarrow 4^+} \sqrt{x-4} = 0$.



2.30 Find δ corresponding to ε for a proof that $\lim_{x \rightarrow 1^-} \sqrt{1-x} = 0$.

We conclude the process of converting our intuitive ideas of various types of limits to rigorous formal definitions by

pursuing a formal definition of infinite limits. To have $\lim_{x \rightarrow a} f(x) = +\infty$, we want the values of the function $f(x)$ to get larger and larger as x approaches a . Instead of the requirement that $|f(x) - L| < \varepsilon$ for arbitrarily small ε when $0 < |x - a| < \delta$ for small enough δ , we want $f(x) > M$ for arbitrarily large positive M when $0 < |x - a| < \delta$ for small enough δ . **Figure 2.43** illustrates this idea by showing the value of δ for successively larger values of M .



In each graph, δ is the smaller of the lengths of the two brown intervals.

Figure 2.43 These graphs plot values of δ for M to show that $\lim_{x \rightarrow a} f(x) = +\infty$.

Definition

Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a . Then, we have an infinite limit

$$\lim_{x \rightarrow a} f(x) = +\infty$$

if for every $M > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) > M$.

Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a . Then, we have a negative infinite limit

$$\lim_{x \rightarrow a} f(x) = -\infty$$

if for every $M > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) < -M$.

2.5 EXERCISES

In the following exercises, write the appropriate $\varepsilon - \delta$ definition for each of the given statements.

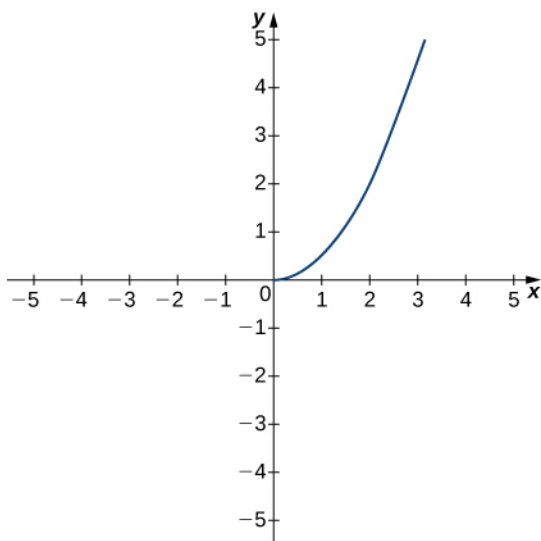
176. $\lim_{x \rightarrow a} f(x) = N$

177. $\lim_{t \rightarrow b} g(t) = M$

178. $\lim_{x \rightarrow c} h(x) = L$

179. $\lim_{x \rightarrow a} \varphi(x) = A$

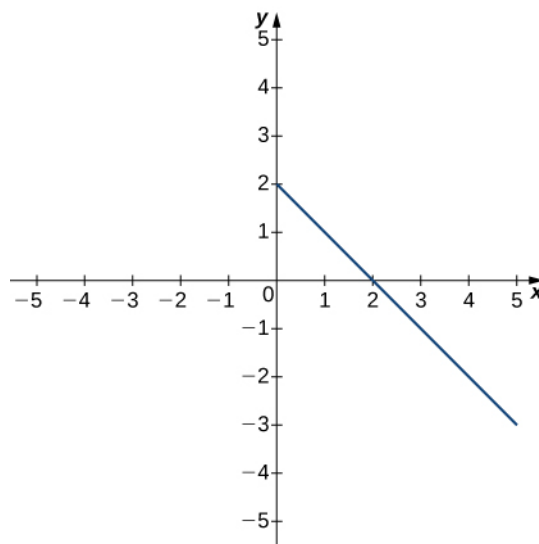
The following graph of the function f satisfies $\lim_{x \rightarrow 2} f(x) = 2$. In the following exercises, determine a value of $\delta > 0$ that satisfies each statement.



180. If $0 < |x - 2| < \delta$, then $|f(x) - 2| < 1$.

181. If $0 < |x - 2| < \delta$, then $|f(x) - 2| < 0.5$.

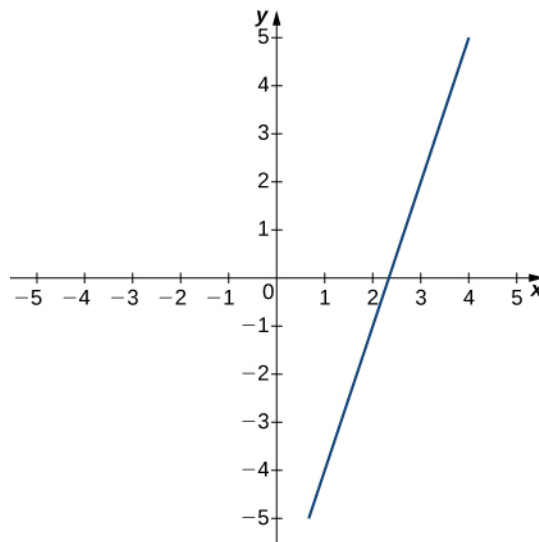
The following graph of the function f satisfies $\lim_{x \rightarrow 3} f(x) = -1$. In the following exercises, determine a value of $\delta > 0$ that satisfies each statement.



182. If $0 < |x - 3| < \delta$, then $|f(x) + 1| < 1$.

183. If $0 < |x - 3| < \delta$, then $|f(x) + 1| < 2$.

The following graph of the function f satisfies $\lim_{x \rightarrow 3} f(x) = 2$. In the following exercises, for each value of ε , find a value of $\delta > 0$ such that the precise definition of limit holds true.



184. $\varepsilon = 1.5$

185. $\varepsilon = 3$

[T] In the following exercises, use a graphing calculator to find a number δ such that the statements hold true.

186. $\left| \sin(2x) - \frac{1}{2} \right| < 0.1$, whenever $\left| x - \frac{\pi}{12} \right| < \delta$

187. $|\sqrt{x-4} - 2| < 0.1$, whenever $|x - 8| < \delta$

In the following exercises, use the precise definition of limit to prove the given limits.

188. $\lim_{x \rightarrow 2} (5x + 8) = 18$

189. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$

190. $\lim_{x \rightarrow 2} \frac{2x^2 - 3x - 2}{x - 2} = 5$

191. $\lim_{x \rightarrow 0} x^4 = 0$

192. $\lim_{x \rightarrow 2} (x^2 + 2x) = 8$

In the following exercises, use the precise definition of limit to prove the given one-sided limits.

193. $\lim_{x \rightarrow 5^-} \sqrt{5 - x} = 0$

194.

$\lim_{x \rightarrow 0^+} f(x) = -2$, where $f(x) = \begin{cases} 8x - 3, & \text{if } x < 0 \\ 4x - 2, & \text{if } x \geq 0 \end{cases}$

195. $\lim_{x \rightarrow 1^-} f(x) = 3$, where $f(x) = \begin{cases} 5x - 2, & \text{if } x < 1 \\ 7x - 1, & \text{if } x \geq 1 \end{cases}$

In the following exercises, use the precise definition of limit to prove the given infinite limits.

196. $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$

197. $\lim_{x \rightarrow -1} \frac{3}{(x+1)^2} = \infty$

198. $\lim_{x \rightarrow 2} -\frac{1}{(x-2)^2} = -\infty$

199. An engineer is using a machine to cut a flat square of Aerogel of area 144 cm^2 . If there is a maximum error tolerance in the area of 8 cm^2 , how accurately must the engineer cut on the side, assuming all sides have the same length? How do these numbers relate to δ , ε , a , and L ?

200. Use the precise definition of limit to prove that the following limit does not exist: $\lim_{x \rightarrow 1} \frac{|x-1|}{x-1}$.

201. Using precise definitions of limits, prove that $\lim_{x \rightarrow 0} f(x)$ does not exist, given that $f(x)$ is the ceiling function. (Hint: Try any $\delta < 1$.)

202. Using precise definitions of limits, prove that $\lim_{x \rightarrow 0} f(x)$ does not exist: $f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$ (Hint: Think about how you can always choose a rational number $0 < r < d$, but $|f(r) - 0| = 1$.)

203. Using precise definitions of limits, determine $\lim_{x \rightarrow 0} f(x)$ for $f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$ (Hint: Break into two cases, x rational and x irrational.)

204. Using the function from the previous exercise, use the precise definition of limits to show that $\lim_{x \rightarrow a} f(x)$ does not exist for $a \neq 0$.

For the following exercises, suppose that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ both exist. Use the precise definition of limits to prove the following limit laws:

205. $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$

206. $\lim_{x \rightarrow a} [cf(x)] = cL$ for any real constant c (Hint: Consider two cases: $c = 0$ and $c \neq 0$.)

207. $\lim_{x \rightarrow a} [f(x)g(x)] = LM$. (Hint: $|f(x)g(x) - LM| = |f(x)g(x) - f(x)M + f(x)M - LM| \leq |f(x)||g(x) - M| + |M||f(x) - L|$.)

CHAPTER 2 REVIEW

KEY TERMS

average velocity the change in an object's position divided by the length of a time period; the average velocity of an object over a time interval $[t, a]$ (if $t < a$ or $[a, t]$ if $t > a$), with a position given by $s(t)$, that is

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}$$

constant multiple law for limits the limit law $\lim_{x \rightarrow a} c f(x) = c \cdot \lim_{x \rightarrow a} f(x) = cL$

continuity at a point A function $f(x)$ is continuous at a point a if and only if the following three conditions are satisfied: (1) $f(a)$ is defined, (2) $\lim_{x \rightarrow a} f(x)$ exists, and (3) $\lim_{x \rightarrow a} f(x) = f(a)$

continuity from the left A function is continuous from the left at b if $\lim_{x \rightarrow b^-} f(x) = f(b)$

continuity from the right A function is continuous from the right at a if $\lim_{x \rightarrow a^+} f(x) = f(a)$

continuity over an interval a function that can be traced with a pencil without lifting the pencil; a function is continuous over an open interval if it is continuous at every point in the interval; a function $f(x)$ is continuous over a closed interval of the form $[a, b]$ if it is continuous at every point in (a, b) , and it is continuous from the right at a and from the left at b

difference law for limits the limit law $\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L - M$

differential calculus the field of calculus concerned with the study of derivatives and their applications

discontinuity at a point A function is discontinuous at a point or has a discontinuity at a point if it is not continuous at the point

epsilon-delta definition of the limit $\lim_{x \rightarrow a} f(x) = L$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$

infinite discontinuity An infinite discontinuity occurs at a point a if $\lim_{x \rightarrow a^-} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^+} f(x) = \pm\infty$

infinite limit A function has an infinite limit at a point a if it either increases or decreases without bound as it approaches a

instantaneous velocity The instantaneous velocity of an object with a position function that is given by $s(t)$ is the value that the average velocities on intervals of the form $[t, a]$ and $[a, t]$ approach as the values of t move closer to a , provided such a value exists

integral calculus the study of integrals and their applications

Intermediate Value Theorem Let f be continuous over a closed bounded interval $[a, b]$; if z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$

intuitive definition of the limit If all values of the function $f(x)$ approach the real number L as the values of x ($x \neq a$) approach a , $f(x)$ approaches L

jump discontinuity A jump discontinuity occurs at a point a if $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist, but

$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$$

limit the process of letting x or t approach a in an expression; the limit of a function $f(x)$ as x approaches a is the value

that $f(x)$ approaches as x approaches a

limit laws the individual properties of limits; for each of the individual laws, let $f(x)$ and $g(x)$ be defined for all $x \neq a$ over some open interval containing a ; assume that L and M are real numbers so that $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$; let c be a constant

multivariable calculus the study of the calculus of functions of two or more variables

one-sided limit A one-sided limit of a function is a limit taken from either the left or the right

power law for limits the limit law $\lim_{x \rightarrow a} (f(x))^n = \left(\lim_{x \rightarrow a} f(x) \right)^n = L^n$ for every positive integer n

product law for limits the limit law $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = L \cdot M$

quotient law for limits the limit law $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{M}$ for $M \neq 0$

removable discontinuity A removable discontinuity occurs at a point a if $f(x)$ is discontinuous at a , but $\lim_{x \rightarrow a} f(x)$ exists

root law for limits the limit law $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L}$ for all L if n is odd and for $L \geq 0$ if n is even

secant A secant line to a function $f(x)$ at a is a line through the point $(a, f(a))$ and another point on the function; the slope of the secant line is given by $m_{\text{sec}} = \frac{f(x) - f(a)}{x - a}$

squeeze theorem states that if $f(x) \leq g(x) \leq h(x)$ for all $x \neq a$ over an open interval containing a and $\lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} h(x)$ where L is a real number, then $\lim_{x \rightarrow a} g(x) = L$

sum law for limits The limit law $\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L + M$

tangent A tangent line to the graph of a function at a point $(a, f(a))$ is the line that secant lines through $(a, f(a))$ approach as they are taken through points on the function with x -values that approach a ; the slope of the tangent line to a graph at a measures the rate of change of the function at a

triangle inequality If a and b are any real numbers, then $|a + b| \leq |a| + |b|$

vertical asymptote A function has a vertical asymptote at $x = a$ if the limit as x approaches a from the right or left is infinite

KEY EQUATIONS

- **Slope of a Secant Line**

$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a}$$

- **Average Velocity over Interval** $[a, t]$

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}$$

- **Intuitive Definition of the Limit**

$$\lim_{x \rightarrow a} f(x) = L$$

- **Two Important Limits**

$$\lim_{x \rightarrow a} x = a \quad \lim_{x \rightarrow a} c = c$$

- **One-Sided Limits**

$$\lim_{x \rightarrow a^-} f(x) = L \quad \lim_{x \rightarrow a^+} f(x) = L$$

- **Infinite Limits from the Left**

$$\lim_{x \rightarrow a^-} f(x) = +\infty \quad \lim_{x \rightarrow a^-} f(x) = -\infty$$

- **Infinite Limits from the Right**

$$\lim_{x \rightarrow a^+} f(x) = +\infty \quad \lim_{x \rightarrow a^+} f(x) = -\infty$$

- **Two-Sided Infinite Limits**

$$\lim_{x \rightarrow a} f(x) = +\infty : \lim_{x \rightarrow a^-} f(x) = +\infty \text{ and } \lim_{x \rightarrow a^+} f(x) = +\infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty : \lim_{x \rightarrow a^-} f(x) = -\infty \text{ and } \lim_{x \rightarrow a^+} f(x) = -\infty$$

- **Basic Limit Results**

$$\lim_{x \rightarrow a} x = a \quad \lim_{x \rightarrow a} c = c$$

- **Important Limits**

$$\lim_{\theta \rightarrow 0} \sin \theta = 0$$

$$\lim_{\theta \rightarrow 0} \cos \theta = 1$$

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$$

KEY CONCEPTS

2.1 A Preview of Calculus

- Differential calculus arose from trying to solve the problem of determining the slope of a line tangent to a curve at a point. The slope of the tangent line indicates the rate of change of the function, also called the *derivative*. Calculating a derivative requires finding a limit.
- Integral calculus arose from trying to solve the problem of finding the area of a region between the graph of a function and the x -axis. We can approximate the area by dividing it into thin rectangles and summing the areas of these rectangles. This summation leads to the value of a function called the *integral*. The integral is also calculated by finding a limit and, in fact, is related to the derivative of a function.
- Multivariable calculus enables us to solve problems in three-dimensional space, including determining motion in space and finding volumes of solids.

2.2 The Limit of a Function

- A table of values or graph may be used to estimate a limit.
- If the limit of a function at a point does not exist, it is still possible that the limits from the left and right at that point may exist.
- If the limits of a function from the left and right exist and are equal, then the limit of the function is that common value.
- We may use limits to describe infinite behavior of a function at a point.

2.3 The Limit Laws

- The limit laws allow us to evaluate limits of functions without having to go through step-by-step processes each time.
- For polynomials and rational functions, $\lim_{x \rightarrow a} f(x) = f(a)$.

- You can evaluate the limit of a function by factoring and canceling, by multiplying by a conjugate, or by simplifying a complex fraction.
- The squeeze theorem allows you to find the limit of a function if the function is always greater than one function and less than another function with limits that are known.

2.4 Continuity

- For a function to be continuous at a point, it must be defined at that point, its limit must exist at the point, and the value of the function at that point must equal the value of the limit at that point.
- Discontinuities may be classified as removable, jump, or infinite.
- A function is continuous over an open interval if it is continuous at every point in the interval. It is continuous over a closed interval if it is continuous at every point in its interior and is continuous at its endpoints.
- The composite function theorem states: If $f(x)$ is continuous at L and $\lim_{x \rightarrow a} g(x) = L$, then $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x)) = f(L)$.
- The Intermediate Value Theorem guarantees that if a function is continuous over a closed interval, then the function takes on every value between the values at its endpoints.

2.5 The Precise Definition of a Limit

- The intuitive notion of a limit may be converted into a rigorous mathematical definition known as the *epsilon-delta definition of the limit*.
- The epsilon-delta definition may be used to prove statements about limits.
- The epsilon-delta definition of a limit may be modified to define one-sided limits.

CHAPTER 2 REVIEW EXERCISES

True or False. In the following exercises, justify your answer with a proof or a counterexample.

208. A function has to be continuous at $x = a$ if the $\lim_{x \rightarrow a} f(x)$ exists.

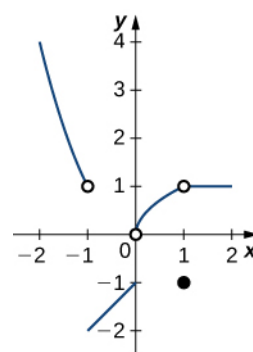
209. You can use the quotient rule to evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

210. If there is a vertical asymptote at $x = a$ for the function $f(x)$, then f is undefined at the point $x = a$.

211. If $\lim_{x \rightarrow a} f(x)$ does not exist, then f is undefined at the point $x = a$.

212. Using the graph, find each limit or explain why the limit does not exist.

- $\lim_{x \rightarrow -1} f(x)$
- $\lim_{x \rightarrow 1} f(x)$
- $\lim_{x \rightarrow 0^+} f(x)$
- $\lim_{x \rightarrow 2} f(x)$



In the following exercises, evaluate the limit algebraically or explain why the limit does not exist.

213. $\lim_{x \rightarrow 2} \frac{2x^2 - 3x - 2}{x - 2}$

$$214. \lim_{x \rightarrow 0} 3x^2 - 2x + 4$$

$$215. \lim_{x \rightarrow 3} \frac{x^3 - 2x^2 - 1}{3x - 2}$$

$$216. \lim_{x \rightarrow \pi/2} \frac{\cot x}{\cos x}$$

$$217. \lim_{x \rightarrow -5} \frac{x^2 + 25}{x + 5}$$

$$218. \lim_{x \rightarrow 2} \frac{3x^2 - 2x - 8}{x^2 - 4}$$

$$219. \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1}$$

$$220. \lim_{x \rightarrow 1} \frac{x^2 - 1}{\sqrt{x} - 1}$$

$$221. \lim_{x \rightarrow 4} \frac{4 - x}{\sqrt{x} - 2}$$

$$222. \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} - 2}$$

In the following exercises, use the squeeze theorem to prove the limit.

$$223. \lim_{x \rightarrow 0} x^2 \cos(2\pi x) = 0$$

$$224. \lim_{x \rightarrow 0} x^3 \sin\left(\frac{\pi}{x}\right) = 0$$

225. Determine the domain such that the function $f(x) = \sqrt{x - 2} + xe^x$ is continuous over its domain.

In the following exercises, determine the value of c such that the function remains continuous. Draw your resulting function to ensure it is continuous.

$$226. f(x) = \begin{cases} x^2 + 1, & x > c \\ 2x, & x \leq c \end{cases}$$

$$227. f(x) = \begin{cases} \sqrt{x + 1}, & x > -1 \\ x^2 + c, & x \leq -1 \end{cases}$$

In the following exercises, use the precise definition of limit to prove the limit.

$$228. \lim_{x \rightarrow 1} (8x + 16) = 24$$

$$229. \lim_{x \rightarrow 0} x^3 = 0$$

230. A ball is thrown into the air and the vertical position is given by $x(t) = -4.9t^2 + 25t + 5$. Use the Intermediate Value Theorem to show that the ball must land on the ground sometime between 5 sec and 6 sec after the throw.

231. A particle moving along a line has a displacement according to the function $x(t) = t^2 - 2t + 4$, where x is measured in meters and t is measured in seconds. Find the average velocity over the time period $t = [0, 2]$.

232. From the previous exercises, estimate the instantaneous velocity at $t = 2$ by checking the average velocity within $t = 0.01$ sec.